

Assignment #1

Marks: 64

Section A (5 × 2 = 10 Marks)

Q1. [CO1] What are the features of Data Warehousing?

Answer:

Data Warehousing is a centralized repository of integrated data from multiple sources. Its main features are:

1. Subject-Oriented:

- Data is organized around key subjects such as customers, sales, products rather than specific applications.
- *Example:* A retail warehouse stores data by product categories or sales regions rather than by daily transactions.

2. Integrated:

- Data from multiple heterogeneous sources is cleaned, transformed, and stored in a consistent format.
- *Example:* Combining data from SQL databases, Excel files, and CRM systems.

3. Time-Variant:

- Historical data is maintained to allow trend analysis over time.
- *Example:* Sales trends over the last 5 years.

4. Non-Volatile:

- Once entered into the warehouse, data is stable and read-only, primarily for querying and analysis.
- *Example:* Historical sales data is not changed by daily operations.

5. Support for Decision-Making:

- Optimized for analytical queries and business intelligence rather than transaction processing.
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Q2. [CO1] What is the difference between Data Mining and Data Warehousing?

Feature	Data Warehousing	Data Mining
Purpose	Store large volumes of integrated data	Extract hidden patterns and knowledge from data

Feature	Data Warehousing	Data Mining
Data Orientation	Focus on historical and consolidated data	Focus on discovering knowledge or trends
Process	ETL (Extract, Transform, Load) process	Uses algorithms like classification, clustering, association
Output	Provides organized data for analysis	Provides insights, patterns, and predictions
Example	A warehouse storing all company sales	Analyzing sales data to predict customer buying patterns

Q3. [CO1] What are the benefits of Data Mining?

Answer:

Data Mining provides several benefits to organizations:

1. **Business Decision Support:** Helps managers make informed decisions based on patterns.
2. **Market Analysis:** Identifies customer buying habits and market trends.
3. **Fraud Detection:** Detects unusual patterns in transactions.
4. **Customer Retention:** Predicts churn and helps in personalized marketing.
5. **Cost Reduction:** Improves efficiency by identifying redundant processes.

Example: Banks use data mining to detect fraudulent credit card transactions.

Q4. [CO1] What is Data Mining?

Answer:

Data Mining is the process of **discovering hidden patterns, relationships, and knowledge from large volumes of data.**

- It involves analyzing historical data and extracting actionable insights using techniques like classification, clustering, and association.

Example: Online retailers use data mining to recommend products based on previous purchases.

Q5. [CO1] What type of information is collected in Data Mining?

Answer:

Data mining collects **various types of information**, including:

1. **Transactional Data:** Purchases, sales, and order history.
2. **Customer Data:** Demographics, preferences, behavior patterns.

3. **Temporal Data:** Time-based information like seasonal sales trends.
 4. **Spatial Data:** Location-based information.
 5. **Text Data:** Customer reviews, feedback, and social media data.
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Section B (7 × 3 = 21 Marks)

Q1. [CO1] Discuss all steps of Data Warehouse process.

Answer:

The data warehouse process involves several key steps:

1. **Data Source Identification:**
 - Identify and select relevant sources like databases, spreadsheets, and logs.
 2. **Data Extraction:**
 - Extract data from heterogeneous sources.
 3. **Data Cleaning:**
 - Remove inconsistencies, duplicates, and missing values.
 - *Example:* Correct misspelled customer names.
 4. **Data Transformation:**
 - Convert data into a common format for analysis.
 - *Example:* Convert all date formats to YYYY-MM-DD.
 5. **Data Loading:**
 - Load transformed data into the warehouse.
 6. **Data Storage:**
 - Store data in structured formats (fact and dimension tables).
 7. **Data Access & Analysis:**
 - Provide tools for querying, reporting, and OLAP operations.
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Q2. [CO1] What kinds of data can be mined?

Answer:

Data mining can be performed on various types of data:

1. **Relational Databases:** Structured data in tables.
2. **Data Warehouses:** Consolidated historical data.
3. **Transactional Data:** Daily business operations.
4. **Spatial and Temporal Data:** Geographic and time-based data.

5. **Multimedia Data:** Images, videos, audio.
 6. **Textual Data:** Emails, web pages, documents.
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Q3. [CO1] Discuss Functionality of Data Mining.

Answer:

Data Mining has the following main functionalities:

1. **Classification:** Assign items to predefined categories.
 - *Example:* Classifying emails as spam or non-spam.
 2. **Clustering:** Group similar items together.
 - *Example:* Segmenting customers based on buying behavior.
 3. **Association Rule Mining:** Discover relationships between variables.
 - *Example:* Customers buying bread often buy butter.
 4. **Regression:** Predict numerical values.
 - *Example:* Predicting house prices based on features.
 5. **Anomaly Detection:** Identify unusual data points.
 - *Example:* Detecting fraudulent bank transactions.
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Section C (11 × 3 = 33 Marks)

Q1. [CO1] Define KDD. Identify and describe phases in KDD Process.

Answer:

KDD (Knowledge Discovery in Databases) is the process of identifying valid, novel, useful, and understandable patterns in data.

Phases of KDD Process:

1. **Data Selection:** Choose relevant data from databases.
2. **Data Preprocessing:** Clean and prepare data by handling missing values and noise.
3. **Data Transformation:** Convert data into appropriate formats for mining.
4. **Data Mining:** Apply algorithms to extract patterns and relationships.
5. **Pattern Evaluation:** Identify truly interesting patterns using evaluation measures.
6. **Knowledge Presentation:** Present findings using visualization and reporting tools.

Example: An e-commerce company uses KDD to discover which products are often bought together.

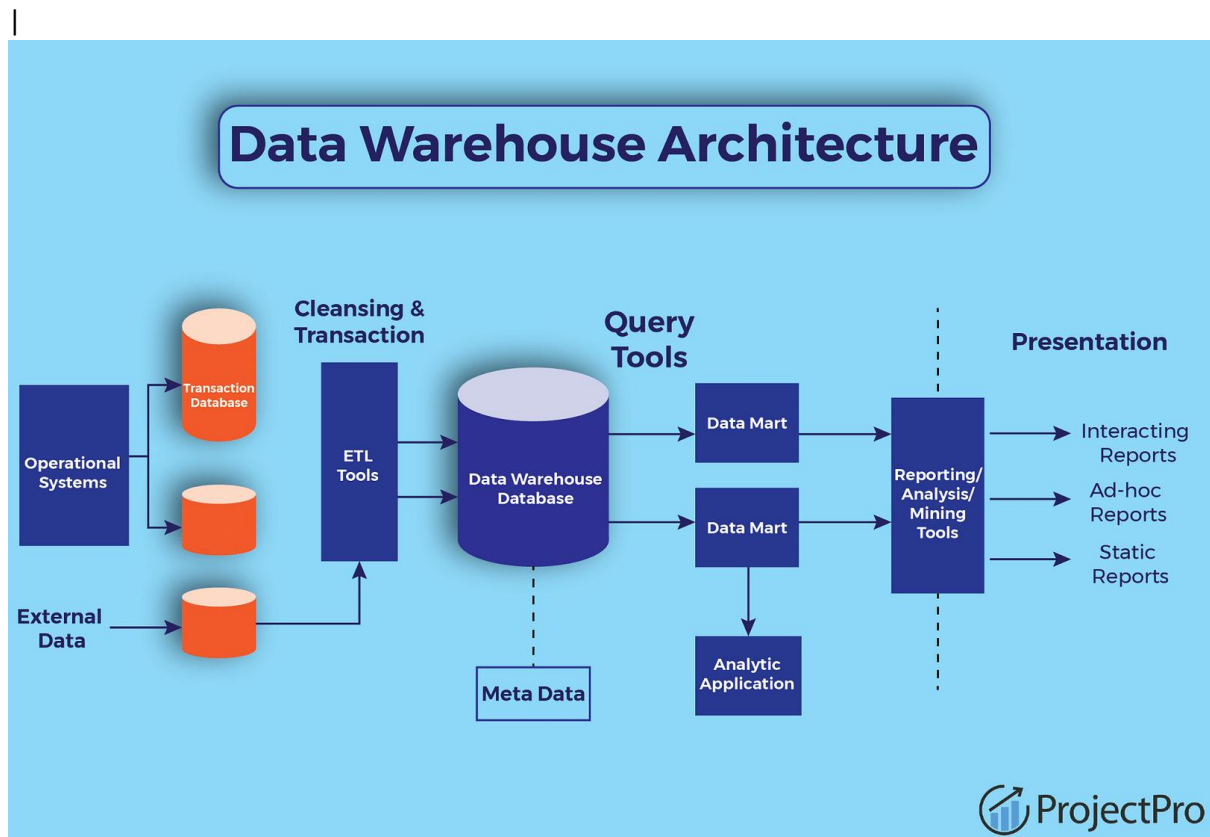
Q2. [CO1] Discuss Architecture of Data Warehouse with diagram.

Answer:

Data Warehouse Architecture typically has three main components:

1. **Data Sources:**
 - Internal (ERP, CRM) and external (market data).
2. **ETL (Extract, Transform, Load) Process:**
 - Extract data → Clean & Transform → Load into warehouse.
3. **Data Warehouse Storage:**
 - Fact tables and dimension tables stored in OLAP cubes.
4. **Presentation Layer:**
 - Tools for reporting, dashboards, and analytics.

Diagram:



Example: Sales and marketing departments analyze consolidated sales data for decision making.

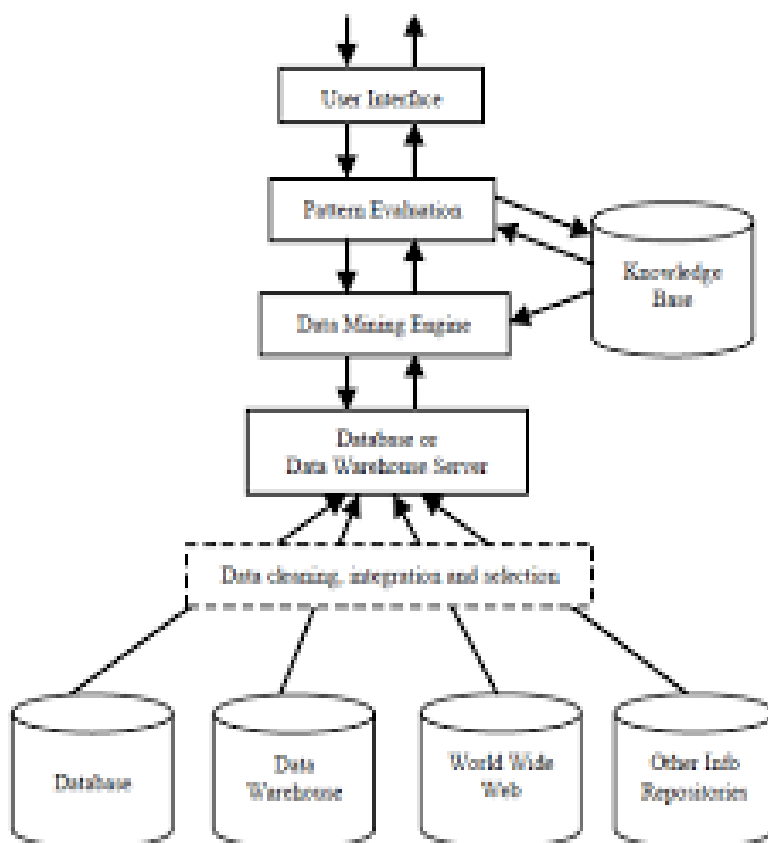
Q3. [CO1] Discuss Architecture of Data Mining System

Answer:

Data Mining System Architecture generally includes the following components:

1. **Database/Data Warehouse:**
 - Source of historical and integrated data.
2. **Data Mining Engine:**
 - Core module that performs classification, clustering, regression, and association.
3. **Pattern Evaluation Module:**
 - Measures the interestingness and validity of discovered patterns.
4. **User Interface:**
 - Enables users to interact, query, and visualize results.
5. **Knowledge Base:**
 - Contains domain knowledge to guide the mining process.

Diagram:



Example: A bank's data mining system predicts loan defaults using historical data and provides insights to the credit team.